AQ9.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Let $f(x, y, z) = (xy, x + y + z^2)$f(x, y, z) = (xy, x + y + z2), where $x, y, z \in R$x, y, z ∈ R. Choose the set of correct options.

- [ ] The domain of ff is R^2R2 and the co-domain of ff is R^2R2
- [ ] The domain of ff is R^3R3 and the co-domain of ff is R^2R2
- [ ] ff is continuous everywhere in its domain
- [ ] ff is not continuous anywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

The domain of ff is R^3R3 and the co-domain of ff is R^2R2

ff is continuous everywhere in its domain

1 point

Let $f(x, y) = \{\frac{\cos y \sin x}{x} \; x \neq 0 \; \cos y \; x = 0$f(x, y) = {xcos y sin xcos yx = 0x = 0 .

Choose the correct options.

- [ ] ff is not defined at $(0, 0)$(0, 0)
- [ ] ff is defined at $(0, 0)$(0, 0)
- [ ] ff is continuous at $(0, 0)$(0, 0)
- [ ] ff is not continuous at $(0, 0)$(0, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

ff is defined at $(0, 0)$(0, 0)

ff is continuous at $(0, 0)$(0, 0)

1 point

Let $h(u, v) = \sin(uv)$, $g(x) = x^3$, $f(y) = \cos y$h(u, v) = sin(uv), g(x) = x3, f(y) = cos y. Let $p(x, y) = h(g(x), f(y))$p(x, y) = h(g(x), f(y))

Choose the correct option(s).

- [ ] pp is a scalar-valued function
- [ ] pp is a vector-valued function

pp is continuous everywhere since $g.f$g.f is continuous everywhere in its domain

☐

pp is continuous everywhere, gg is also continuous everywhere while ff is not continuous everywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

pp is a scalar-valued function

pp is continuous everywhere since $g.f$g.f is continuous everywhere in its domain

1 point

Let $p(t) = (ln(1-t), \frac{1}{t}, 3t)$p(t) = (ln(1 − t), t1, 3t).

Choose the incorrect option(s).

☐

pp is continuous for all real tt less than 11 but not equal to zero

☐

pp is continuous on RR

☐

pp is continuous in $\{t \in R | t < 1\}$$\{t \in R | t < 1\}$

☐

pp is continuous in $\{t \in R | t \neq 1, t \neq 0\}$$\{t \in R | t \quad = 1, t \quad = 0\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

pp is continuous on RR

pp is continuous in $\{t \in R | t < 1\}$$\{t \in R | t < 1\}$

pp is continuous in $\{t \in R | t \neq 1, t \neq 0\}$$\{t \in R | t \quad = 1, t \quad = 0\}$

1 point

Let $f(x, y) = ln(x^2 + y^2 - 1)$f(x, y) = ln(x2 + y2 − 1). Identify where ff is continuous. Note: Equation of a circle with radius rr and center (a, b)(a, b) is given by $(x-a)^2 + (y-b)^2 = r^2$ (x − a)2 + (y − b)2 = r2

○

Everywhere in R^2R2

○

Everywhere in $R^2 \setminus \{(1, 0), (0, 1)\}$R2 \ {(1, 0), (0, 1)}

○

Everywhere in R^2R2 outside the circle with radius 11 and center (0, 0)(0, 0)

○ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Everywhere in R^2R2 outside the circle with radius 11 and center (0, 0)(0, 0)

Level 2

1 point

Let $g(u, v, w) = \frac{uv}{w}$g(u, v, w) = wuv and $f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$ f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))

where $P(x, y, z) = e^{x^2 + y}$, $Q(x, y, z) = \sqrt{y^2 + z^2 + 3}$, $R(x, y, z) = \sin(xyz) + 5$

P(x, y, z) = ex2+y, Q(x, y, z) = y2 + z2 + 3

√

, R(x, y, z) = sin(xyz) + 5.

Choose the correct option(s).

- [ ] f f is a vector-valued function
- [ ] f f is a scalar-valued function
- [ ] f f is continuous everywhere in its domain.
- [ ] P P is continuous everywhere in its domain, while Q Q is not continuous everywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

f f is a scalar-valued function

f f is continuous everywhere in its domain.

1 point

Let

$f: R^2 \rightarrow R, f(x, y) = \{\frac{x^2}{x^2 + y^2}(x, y) \neq (0, 0)$ f : R2 → R, f(x, y) = {x2+y2x20(x, y) = (0, 0)(x, y) =

$0 \quad (x, y) = (0, 0)$

(0, 0).

Choose the correct option(s).

- [ ] $\lim_{(x, y) \rightarrow (0, 0)} f$ lim(x,y)→(0,0) f does not exist
- [ ] $\lim_{(x, y) \rightarrow (0, 0)} f$ lim(x,y)→(0,0) f exists
- [ ] f f is continuous at (0, 0)(0, 0)
- [ ] f f is not continuous at (0, 0)(0, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x, y) \rightarrow (0, 0)} f$ lim(x,y)→(0,0) f does not exist

f f is not continuous at (0, 0)(0, 0)

1 point

Let

$f: R^2 \rightarrow R, f(x, y) = \{\frac{xy}{x^2 + y^2}(x, y) \neq (0, 0)$ f : R2 → R, f(x, y) = {x2+y2xy0(x, y) = (0, 0)(x, y) =

$0 \quad (x, y) = (0, 0)$

(0, 0).

Choose the correct option(s).

- [ ] $\lim_{(x, y) \rightarrow (0, 0)} f$ lim(x,y)→(0,0) f does not exist

☐
$\lim_{(x,\, y)\, \to\, (0,0)} f$ lim(x,y)→(0,0) f exists

☐
f f is continuous at $(0,0)$(0, 0)

☐
f f is not continuous at $(0,0)$(0, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\lim_{(x,\, y)\, \to\, (0,0)} f$ lim(x,y)→(0,0) f does not exist
f f is not continuous at $(0,0)$(0, 0)

1 point

Let $f(r) = \frac{1}{<r,r>-1}$, where $r = (x, y, z)$ and the inner product $<,>$ is the dot product
Choose the set of incorrect options.

☐
f f is continuous throughout R^3 R3

☐
f f is continuous throughout $R^3 \setminus \{(0,0,0)\}$R3 \ {(0, 0, 0)}

☐
f f is discontinuous on the unit sphere $\{r \in R^3 : \| r \| = 1\}$ {r ∈ R3 : ‖r‖ = 1}

☐
f f is continuous throughout $R^3 \setminus \{(1,0,0)\}$R3 \ {(1, 0, 0)}

No, the answer is incorrect.
Score: 0
Accepted Answers:
f f is continuous throughout R^3 R3
f f is continuous throughout $R^3 \setminus \{(0,0,0)\}$R3 \ {(0, 0, 0)}
f f is continuous throughout $R^3 \setminus \{(1,0,0)\}$R3 \ {(1, 0, 0)}

Let $f : R^2 \to R, f(x, y)$f : R2 → R, f(x, y)= $\{\frac{x^k y}{x^4 + y^4}(x, y) \neq (0,0)$ {x4+y4xky0(x, y) = (0, 0)(x, y) = $0 \quad (x, y) = (0,0)$

(0, 0). Find the minimum possible positive integer value of kk such that f f is continuous at (0,0).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

Check Answers
Your score is: 0/10